

KSAT Ranging Portal

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Welcome to ranging!

This page introduces the radiometric products that KSAT offers on select antennas in our network. Radiometrics includes:

range measurements

range rate (aka Doppler)

integrated carrier phase (aka integrated Doppler, accumulated delta range)

 Support for integrated carrier phase is in development.

Supporting products include:

angle data from the antenna controller

signal metrics, such as signal power and signal/noise ratio

atmospheric measurements to support troposphere models

Data products are delivered to the user in the form of [CCSDS 503.0-B-2](#) TDM files.

Ranging operations are controlled using API calls, allowing the user to reconfigure ranging during a pass, and to collect the ranging data.

For information about the ranging API, see:

For information about TDM files, see:

- [Overview](#): Overview of the TDM format used for KSAT ranging products.
- [TRACK](#): Radiometric tracking data (range, carrier phase, range rate):
- [ANGLE](#): Antenna pointing angles
- [SIGMET](#): Signal metrics
- [METEO](#): Meteorology

For information about how to process the data in the TDM files, see:

- [ranging model](#)
- [delay model](#)
- [ranging detailed algorithm](#)
- [range rate model](#)